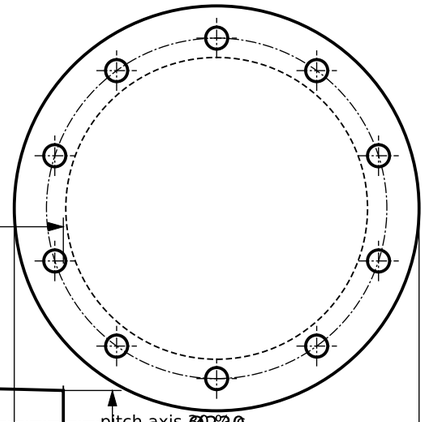
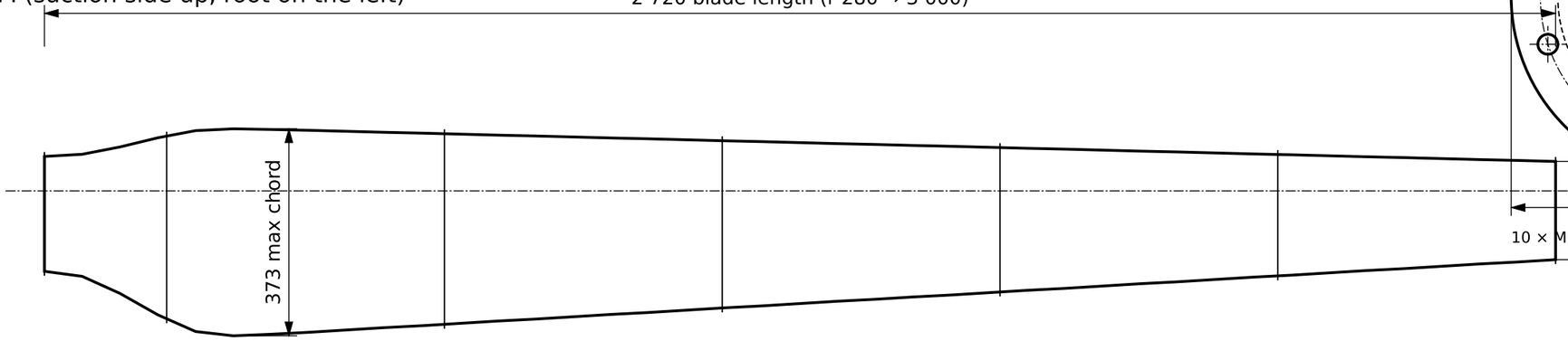
ROOT END (from the hub)



PLANFORM (suction side up, root on the left)



r mm	chord	twist °	t/c	cap mm
280	207	20.0	1.00	4.25
400	254	20.0	0.71	4.25
600	370	20.0	0.32	4.25
800	360	17.1	0.28	4.25
1000	343	13.4	0.26	4.10
1400	310	8.5	0.24	3.38
1800	277	5.5	0.21	2.41
2200	244	3.6	0.19	1.44
2600	210	2.2	0.18	1.00
3000	177	1.2	0.18	1.00

AXONOMETRIC VIEW (not to scale)



NOTES

1. Aerofoil family and twist from analysis/rotor/bem.py (TSR 5.5, cl 1.0, R 3.0 m, root r 0.28 m); twist given at the pitch axis, positive nose up towards feather; blade set at −2° in the arc slots of the hub root flange (NS6-130).
2. Structure doc 30 §2 / doc 10 model: two glass UD caps (table), ±45° glass skins, PVC foam core, single shear web 1.5 mm glass; root wall builds up 12 mm over the first 0.7 m. Cap buckling at minimum thickness governs (use 1.00); strength 0.43 (SLM H), fatigue 0.10 with simulated loads (doc 31).
3. Root: 10 bonded steel inserts M12 NS6-105, PCD 185, pull-out ≥ 60 kN each (test). Bolt force 43 kN at SLM H. Sections and laminate: NS6-101; mould NS6-110.
4. Mass 20.6 kg ± 0.3 kg; static moment matched within 0.5 % per set of three (balance class doc 09). Erosion tape on the outer 1 m of the leading edge.
5. First flap 15.2 Hz, first edge 38.3 Hz at standstill. Tip deflection 0.06 m, tower clearance 0.83 m.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BLADE · PLANFORM, TWIST AND ROOT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-100	B · 2026-11-06	1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
E-glass/epoxy infused; glass UD caps; PVC core shear web	20.6 kg	Claude / not checked